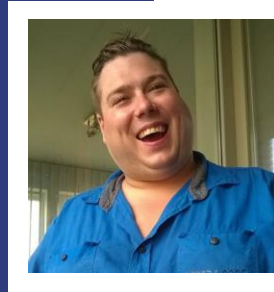




#XamExpertDay

Improving people with disabilities' life With Xamarin.Essentia ls



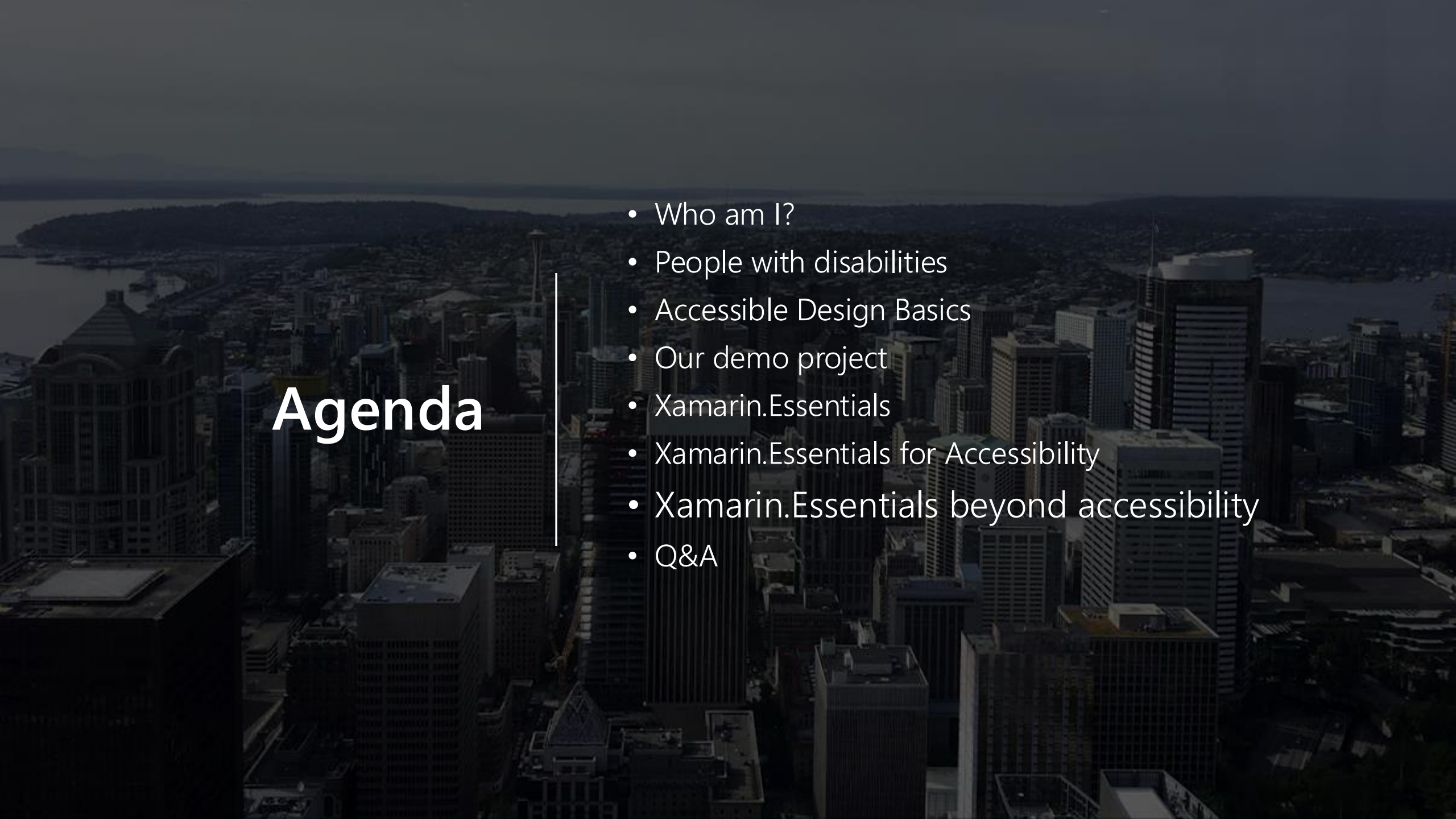
Dennie Declercq



[@DennieDeclercq](https://twitter.com/DennieDeclercq)



Intro

An aerial, high-angle view of a city skyline, likely Seattle, with the Space Needle visible in the distance. The image is dark and serves as a background for the text.

Agenda

- Who am I?
- People with disabilities
- Accessible Design Basics
- Our demo project
- Xamarin.Essentials
- Xamarin.Essentials for Accessibility
- Xamarin.Essentials beyond accessibility
- Q&A

About me

- Dennie Declercq
- Microsoft MVP
- DDSOFT nonprofit (Belgium)
- Autism
- @DennieDeclercq
- #DreamingIsBelieving



♥️#7ThYearMagic 🦋



People With Disabilities

People!

- PWD are People
- People
- Thread human
- Help Human



Kind of disabilities



Intellectual /
Cognitive



Autism



Visual



Auditory



Motoric



Much more...

Cognitive / Intellectual Disability?

Significant limitations in both:

- Intellectual functioning
- Adaptive behavior

Cognitive / Intellectual Disability?

Intellectual Functioning

Also called intelligence

Mental capacity

IQ score between 70-75

Adaptive Behavior

Conceptual Skills

- Language/literacy
- Money
- Time/numbers

Social Skills

- Social responsibility
- Self-esteem

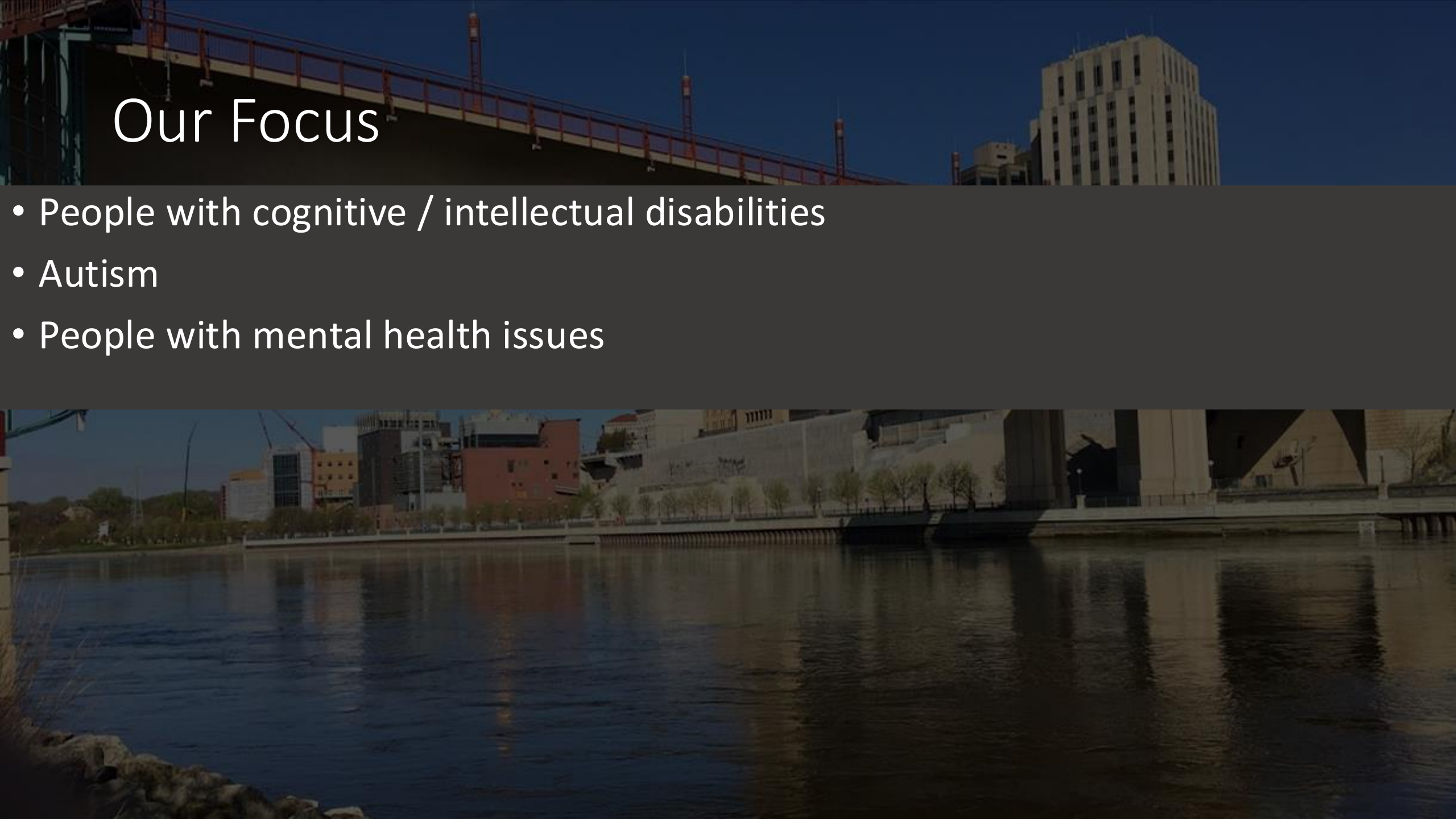
Practical skills

- Activities daily living
- Personal care / healthcare

Principles of Inclusion

- Same activities
- Same district
- Work Together
- Adjustments par person



The background image is a composite of two photographs. The top half shows a close-up of a bridge with a red metal railing and a tall, modern building with many windows in the background under a clear blue sky. The bottom half shows a wide river or canal with a city skyline in the background, including various buildings and a bridge, with the water reflecting the scene.

Our Focus

- People with cognitive / intellectual disabilities
- Autism
- People with mental health issues

Accessible Design Basics

Simplify the layout

- **Less** content per screen
- **Logical** content per screen
- **Light** or white **background**
- A few **accent** colors

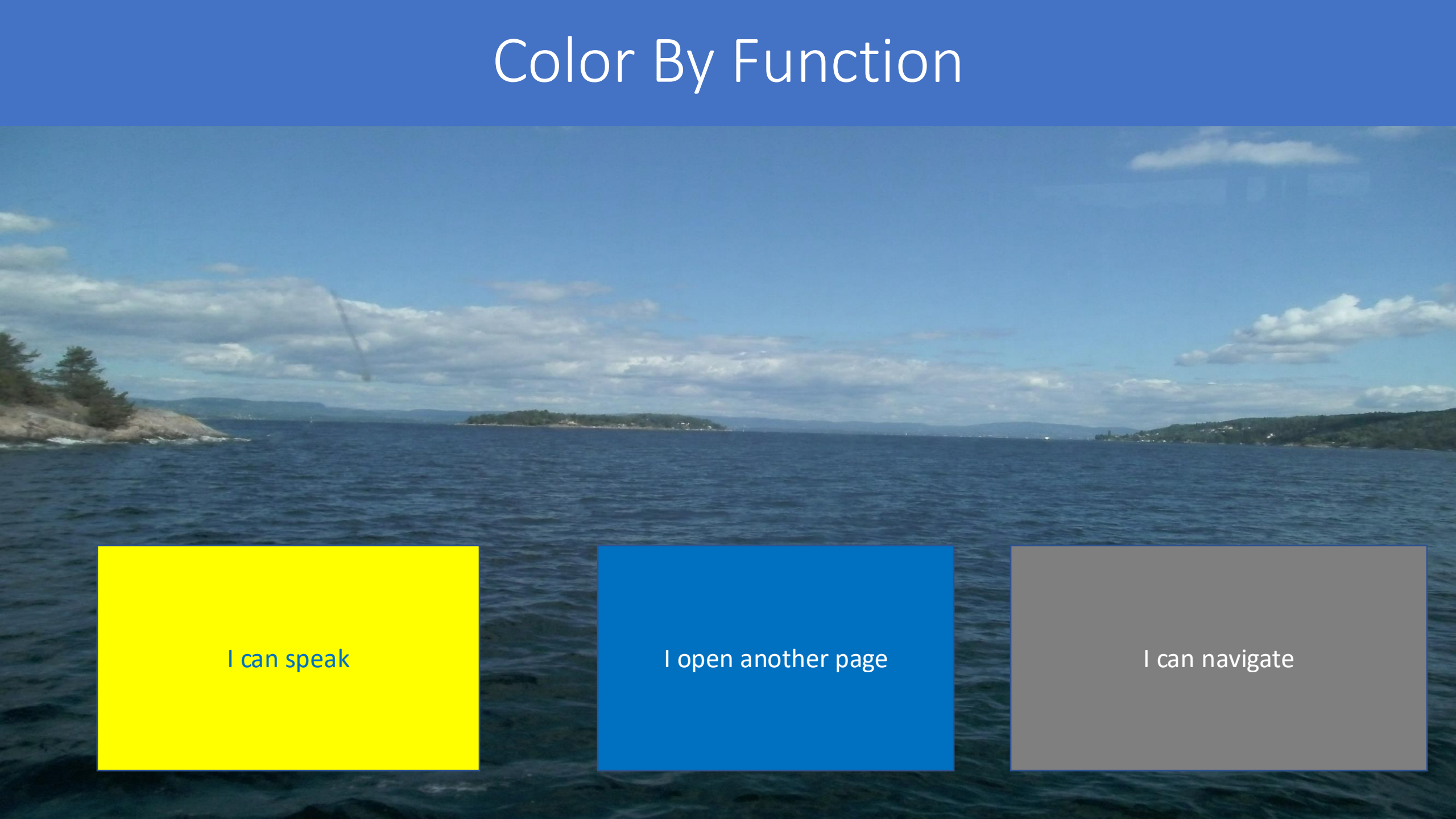
Dark color theme

- Effective: People with Autism

Color Contrast

- Make enough contrast!
- Needed for a lot of people!

Color By Function



I can speak

I open another page

I can navigate

Pictures and Pictograms

- What means the following?

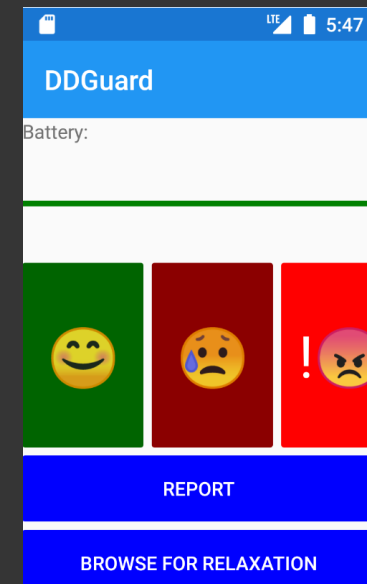
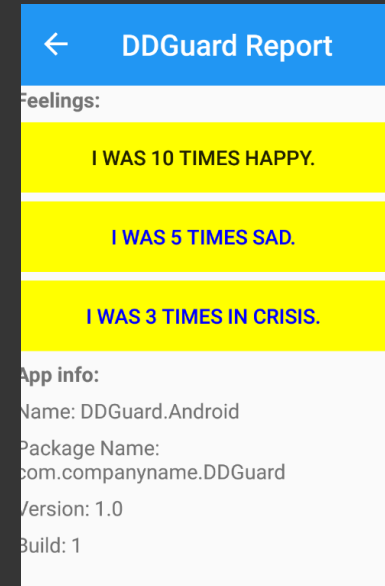
- 酢飯

- The answer is:



Our demo app: 'DDGuard'

- Disability + Mental Health
- Emotion tracker
- Calming down
- GitHub
- (Feel free to contribute)
- <https://github.com/DDSoftDennie/DDGuard>

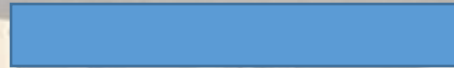
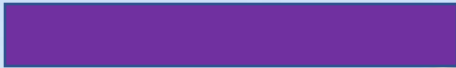


Xamarin.Essentials

Xamarin.Essentials



Windows 10



Shared UI Code

Shared Platform Extensions (Xamarin.Essentials)

Shared C# Backend

Xamarin.Essentials API's

Accelerometer

App Information

Barometer

Battery

Clipboard

Color Converters

Compass

Connectivity

Detect Shake

Device Display Information

Device Information

Email

File System Helpers

Flashlight

Geocoding

Gyroscope

Launcher

Magnetometer

Main Thread

Maps

Open Browser

Orientation Sensor

Phone Dialer

Platform Extensions

Preferences

Secure Storage

Share

SMS

Text-To-Speech

Unit Converters

Version Tracking

Vibrate

Getting Started with Xamarin.Essentials

- Installing NuGet Package
- Using Statement
- Most of the time a single line of code

```
using Xamarin.Essentials
```

Xamarin.Essentials For Accessibility

Text-To-Speech

– Use Case



Reading Difficulties



Complete word-blind

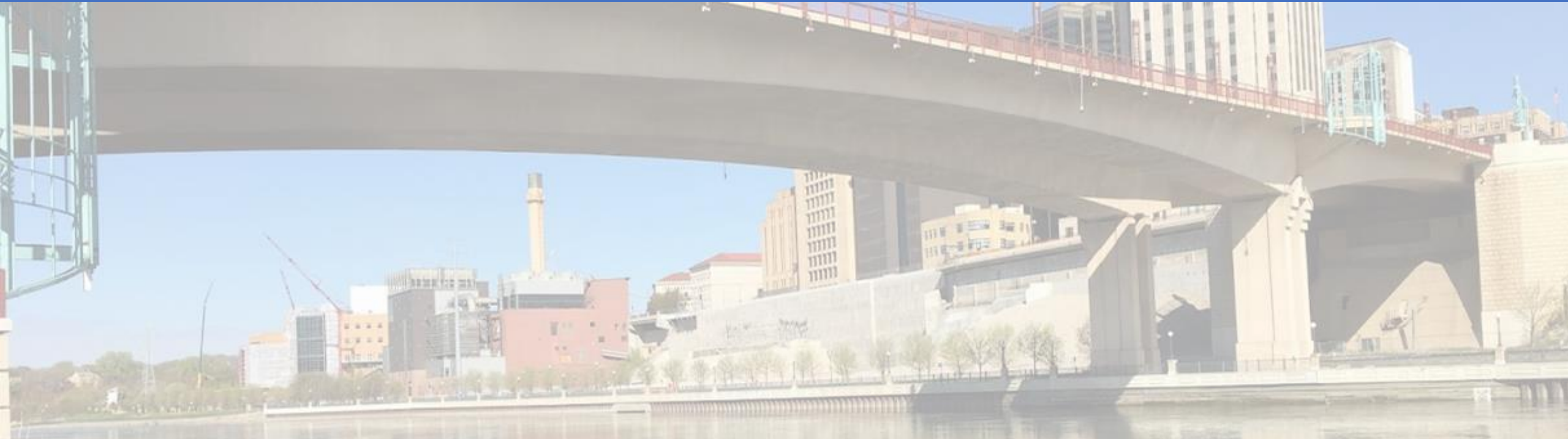


Confirmation



Dyslexia

Text-To-Speech - Code



```
private async void HappyButton_Clicked(object sender, EventArgs e)
{
    //Xamarin.Essentials: Text To Speech
    await TextToSpeech.SpeakAsync(HappyButton.Text);
}
```

Xamarin.Essentials Beyond Accessibility

Preferences: Use Case



Saving Information



Limit Data Usage

Preferences: Code



```
//Xamarin.Essentials: Preferences  
Preferences.Set("Happy", Statics.Happy);  
  
//Xamarin.Essentials: Preferences  
Statics.Happy = Preferences.Get("Happy", 0);
```

Battery: use Case



People forgot loading device



Problems with prediction
battery state



Device Out of functioning in
critic situation

Battery: How to handle?



Visual



With colors



With auditory warning

Battery: Code

```
//Xamarin.Essentials: Battery
//PERMISSIONS: BATTERY_STATS
double dBattery = Battery.ChargeLevel;
BatteryProgressBar.Progress = dBattery;

if(dBattery <= 0.20)
{
    BatteryProgressBar.ProgressColor = Color.Red;
    SpeakBattery();
} else if(dBattery <=0.35)
{
    BatteryProgressBar.ProgressColor = Color.Orange;
} else {
    BatteryProgressBar.ProgressColor = Color.Green;
}
```

Connectivity: Use Case



No Data available



People don't know there's no data connection



App crashes without data



People are confused

Connectivity: Code

```
//Xamarin.Essentials: Connectivity
var connection = Connectivity.NetworkAccess;
if(connection == NetworkAccess.Internet)
{
    NoConnectionButton.IsVisible = false;
    return true;
}
else
{
    NoConnectionButton.IsVisible = true;
    return false;
}
```

Launcher: Use Case



Show your privacy statements



Open online media



Show movies (YouTube)

Launcher: Code



```
if(CheckInternetConnection() == true)
{
    //Xamarin.Essentials: Launcher
    await
Launcher.OpenAsync("https://www.youtube.com/watch?v=nPbWxaF0D-U");
}
```


AppInfo: Use Case



Parents / Caregiver don't know current app info



Lack of knowledge system settings

AppInfo: Code



```
//Xamarin.Essentials: AppInfo
```

```
AppNameLabel.Text += " " + AppInfo.Name;  
AppPackageLabel.Text += " " + AppInfo.PackageName;  
AppVersionLabel.Text += " " + AppInfo.VersionString;  
AppBuildLabel.Text += " " + AppInfo.BuildString;
```

DeviceInfo: Use Case



Same as AppInfo:



Lack of knowledge parent /
caregiver

DeviceInfo: Code



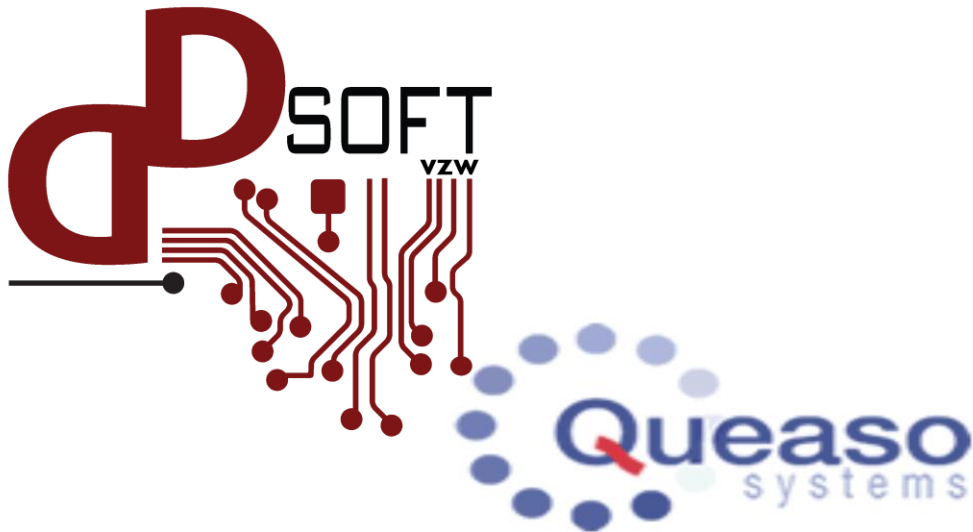
```
//Xamarin.Essentials: DeviceInfo
```

```
DeviceModelLabel.Text += " " + DeviceInfo.Model;  
DeviceManufacturerLabel.Text += " " + DeviceInfo.Manufacturer;  
DevicePlatformLabel.Text += " " + DeviceInfo.Platform;  
DeviceOSLabel.Text += " " + DeviceInfo.VersionString;
```

Watching our demo again (full
attention)

Thanks

- Dennie Declercq
- DDSoft nonprofit (Belgium)
- @DennieDeclercq
- #DreamingIsBelieving



Q & A



Thank You!